

- Q.19 Write a short note on following :
- a) Freon-12 refrigerant (CO4)
 - b) Low temperature insulation (CO4)
- Q.20 What is inert gas and writes any two properties and two use of it. (CO3)
- Q.21 What are the key properties to consider for the selection of refrigerant? (CO4)
- Q.22 What is insulating material and their effect on piping, fitting and valves? (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe the following terms
- a) Carryover (CO1)
 - b) Blower air (CO1)
 - c) Foaming (CO3)
 - d) Thermal insulation (CO4)
- Q.24 Explain construction and working of Natural draft cooling tower in detail. (CO4)
- Q.25 Explain Ion-exchange process for softening of water in detail with the help of neat diagram. (CO4)

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Roll No.

5th Sem/ Chemical, Chemical (Pulp & Paper) Subject : Process Plant Utilities

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The most impure form of water is ____? (CO1)
- a) Rain water b) River water
 - c) Lake water d) Sea Water
- Q.2 Which of the following process is water softening process? (CO2)
- a) Zeolite process
 - b) Lime soda process
 - c) Ion-Exchange process
 - d) All of above
- Q.3 Which type of steam is formed when water is heated above its critical temperature? (CO3)
- a) Saturated steam b) Superheated steam
 - c) Wet steam d) Dry steam

- Q.4 A secondary refrigerant is ____? (CO4)
 a) Carbon dioxide b) Ethylene glycol
 c) Ammonia d) Freons
- Q.5 Safe working property of refrigerant is classified as ____? (CO4)
 a) Freezing point
 b) Thermal conductivity
 c) Flammability
 d) Viscosity
- Q.6 What is the purpose of insulation in a process plant steam system? (CO4)
 a) To reduce heat transfer
 b) To increase heat transfer
 c) To control pressure
 d) To control flow rate

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 °Clarke (°Cl) is the unit of _____. (CO1)
- Q.8 Permanent hardness is due to soluble chlorides and sulphate salt of calcium and magnesium. (True/False) (CO1)

- Q.9 Argon is an inert gas that is commonly used in welding insulation. (True/False) (CO3)
- Q.10 Write the name of any one internal method for water softening. (CO2)
- Q.11 Write the name of any one primary refrigerant. (CO4)
- Q.12 A good insulator material should have _____ thermal conductivity. (Low/high) (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between Temporary hardness and permanent hardness of water. (CO1)
- Q.14 Describe the following (CO3)
 a) Specific volume of steam
 b) Dryness fraction
- Q.15 Write the difference between scale and sludge formation in water boiler. (CO1)
- Q.16 Describe the lime soda process for water softening in brief? (CO2)
- Q.17 Explain hot thermal insulation with example in brief. (CO4)
- Q.18 Explain the reverse osmosis treatment of brackish water. (CO2)